

Vinh Ly

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Summary

Software Engineer with **18 years** of experience designing and operating systems at scale across consumer-facing platforms and enterprise infrastructure. Owned end-to-end delivery of **search ranking infrastructure** serving billions of daily queries at Google and **live marketplace systems** supporting millions of concurrent users at Whatnot. Built and led teams that delivered **distributed data pipelines** processing petabytes daily and **kernel-level optimizations** reducing latency by 40%. Next focus on high-throughput real-time systems and infrastructure reliability at scale. Owns production infrastructure end to end, including on call coverage, deployment automation, and monitoring across distributed services. Comfortable partnering with platform and application teams on reliability work.

Core Skills

Languages and Frameworks: TypeScript, JavaScript, HTML, CSS, SQL

Distributed Systems: large-scale data pipelines, multi-region infrastructure, caching architectures, load balancing, service mesh

Systems Programming: C++, kernel-level development, Windows internals, driver development, low-level optimization

Search & Recommendation: indexing infrastructure, ranking algorithms, relevance scoring, query processing, recommendation systems

Real-Time Systems: stream processing, live video infrastructure, marketplace platforms, fraud detection pipelines, sub-millisecond latency

Work Experience

Staff Software Engineer, YouTube

02/2025 - Present | Remote

- Designing next-generation **video indexing infrastructure** to support 500M+ daily active users with p99 latency under 50ms across global edge network.
- Architecting **real-time analytics pipelines** processing billions of events per day for content recommendation and creator dashboard features.
- Establishing reliability standards and **SLO frameworks** for critical creator-facing services serving multi-million dollar revenue streams.
- Mentoring 4 engineers on **systems design** and **incident response** best practices for high-traffic platform services.
- Leading cross-functional design reviews for **distributed cache invalidation** strategies across 10+ interdependent storage systems.

Staff Software Engineer/Manager, Google

03/2015 - 04/2024 | Remote

- Owned **on-call rotation** for core search serving handling 50K+ queries per second; reduced mean time to resolution from 45 minutes to 12 minutes through improved runbooks and tooling.
- Led team of 8 engineers building **search ranking infrastructure** serving 3.5B queries daily, reducing average query latency from 85ms to 28ms through **kernel-level optimization** of hot paths.
- Built **machine learning feature pipeline** serving 40+ ranking models with sub-second freshness for search relevance improvements that increased clicks by 12%.

- Designed **distributed indexing system** processing 100TB+ daily updates with 99.99% availability, replacing legacy monolith that required weekly maintenance windows.
- Managed technical roadmap and delivery for **relevance infrastructure** across 15 international domains, partnering with ranking and quality teams quarterly.

Software Engineer, Google

02/2011 - 02/2015 | Remote

- Developed **crawling infrastructure** that discovered and processed 20B+ webpages weekly, improving freshness of search index by 35% through distributed fetch scheduling.
- Built internal tooling for **capacity planning** that predicted infrastructure needs 6 months ahead with 92% accuracy, eliminating three outage scenarios.
- Optimized **query log processing** from terabytes daily to actionable signals within 15 minutes using **stream processing** and **MapReduce**.
- Implemented **deduplication pipeline** reducing index size by 18% while preserving query result quality across duplicate detection thresholds.

Software Design Engineer II, Microsoft

10/2007 - 12/2011 | Remote

- Developed **Windows kernel components** for file system minifilter framework, shipping in Windows 8 with 40% performance improvement over previous API surface area.
- Created automated testing framework for **kernel-mode driver validation** that caught 200+ pre-release bugs and reduced customer-facing blue screens by 60%.
- Collaborated with **hardware partner team** to validate storage driver compatibility across 50+ OEM configurations before release cycles.
- Implemented **low-level memory optimization** reducing driver footprint by 25% through profile-guided allocation strategies in **C++**.

Software Design Engineer Intern, Microsoft

06/2006 - 08/2006 | Remote

- Built prototype **Windows Vista search integration** for enterprise file shares, demonstrating 15-second average speedup over existing network search methods.
- Authored internal documentation and **API specifications** for file system filter communication interfaces adopted by three product teams.

Education

B.S. Honors Degrees in Computer Science

Virginia Tech | 2004 - 2007